



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:

[Redacted Roll Number]

Course Name: B.Tech

Subject Name: Basics of Electrical Engineering

Subject Code: 25EL101

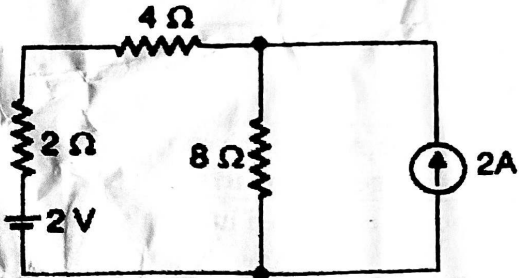
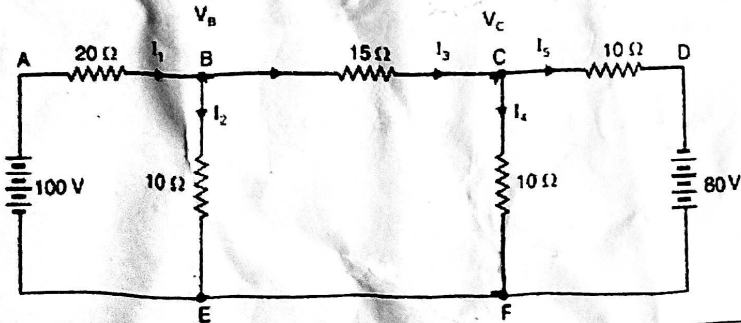
Max.Marks:15

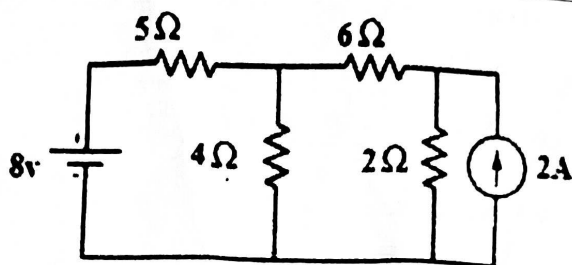
Time: 75 Minutes

Semester: 1st

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions		(1 x 5)		
(a)	Explain the concept of super node in electric circuit.	CO1	K2	2.1.2
(b)	Draw the V-I characteristics for ideal voltage source and ideal current source.	CO1	K2	2.1.2
(c)	Define active and passive elements.	CO1	K2	2.1.2
(d)	Describe true power, reactive power in AC circuit.	CO2	K1	1.1.1
(e)	The current drawn by a pure capacitor of $20 \mu\text{F}$ is 1.382 A from 220 V a.c. supply. Determine the capacitive reactance of the circuit.	CO2	K3	2.1.3
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question		(Same K Levels) (2 x 2)		
(a)	Find the current in 4Ω resistance using mesh analysis. 	CO1	K3	1.2.1 1.4.1
(b)	Determine the current in the various branches of the circuit shown in figure by nodal analysis? 	CO1	K3	1.2.1 1.4.1
(c)	Determine current in 4Ω resistance using Norton's theorem in the following circuit:	CO1	K3	1.2.1 1.4.1

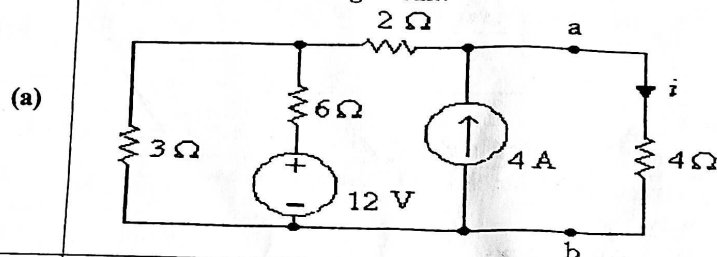


SECTION-C (Minimum Knowledge Level: K3)

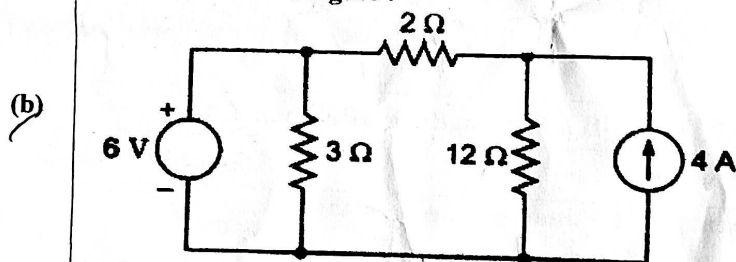
3. Attempt Any one Question

(Same K Levels) (3 x 1)

Determine current in 4Ω resistance using Thevenin's theorem in the following circuit:


CO1 K3 1.2.1
1.4.1

Using superposition theorem, Find the current in 2Ω resistance in below figure :


CO1 K3 1.2.1
1.4.1

4. Attempt Any one Question

(Same K Levels) (3 x 1)

(a) Prove that for a purely capacitive circuit, the supply current leads the supply voltage by a quarter of cycle.

CO2 K2 1.1.1
1.4.1

(b) Prove that the average power consumed in purely inductive circuit is zero.

CO2 K2 1.1.1
1.4.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



Mid Term-II (Session: 2025-26)

Roll No.:

Course Name: B. Tech
Subject Name: Basics of Electrical Engineering
Subject Code: 25EL101
Semester: I

Max. Marks: 15
Time: 75 Minutes

Note:

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3. **Attempt All Questions**
4. **Assume Any Missing Data if required**

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions				(1 x 5)
(a)	Draw the Hysteresis loop showing Residual and Saturation points.	CO3	K2	1.3.1
(b)	Mention any two properties of an Ideal Transformer.	CO3	K2	1.3.1
(c)	State the condition for Maximum Efficiency of a Single Phase Transformer.	CO3	K2	1.3.1 2.3.1
(d)	Write down the relationships between Line Voltages, Phase Voltages, Line current and Phase currents of a Delta connected load.	CO2	K2	2.1.2
(e)	Define Bandwidth of a Series RLC circuit.	CO2	K2	1.3.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question				(Same K Levels) (2 x 2)
(a)	Determine the EMF equation of a Single phase Transformer.	CO3	K2	2.3.1
(b)	Explain and Draw the approximate equivalent circuit of a Single Phase Transformer referred to Primary side.	CO3	K2	1.3.1 2.3.1
(c)	Discuss the analogy between Magnetic and Electrical Circuit.	CO3	K2	1.3.1
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question				(Same K Levels) (3 x 1)
(a)	In a 25 KVA, 2000/200 V transformer the iron losses and copper losses are 350 W and 0.400 KW respectively. Calculate the efficiency at (i) Full load and 0.8 pf lagging (ii) Half load and 0.6 pf lagging.	CO3	K3	2.3.1 1.4.1
(b)	A 20 KVA, 2000/200V, single phase, 50 Hz transformer has a primary resistance of 3.5 ohm and reactance of 4.5 ohm. The secondary resistance and reactance are 0.015 ohm and 0.02 ohm respectively. Determine (i) Equivalent resistance and impedance referred to primary (ii) Total copper loss in the transformer.	CO3	K3	2.3.1 1.4.1



4. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	A series R-L-C circuit consisting of a resistance of 20 ohm, an inductance of 100 mH and a capacitive of 150 μ F is connected across a 230 V, 50 Hz source, Calculate (i) Impedance (ii) Current (iii) Power consumed.	CO2	K3	1.4.1
(b)	Two impedances $Z_1 = (6-j8)$ ohms and $Z_2 = (16+j12)$ ohms are connected in parallel across 230V supply. Determine (i) Supply Current (ii) Power factor of the circuit (iii) Power consumed.	CO2	K3	2.1.2 2.1.3

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Pre - End Term Exam (Session: 2025-26)

Admission No

2

Max. Marks: 20

Time: 90 Minutes

Semester: First

Course Name: B.Tech

Subject Name: Basics of Electrical Engg.

Subject Code: 25EL101

Note:

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SECTION-A (Knowledge Level: K2 & K3)

Q. No.	Question	CO	KL	PI
Attempt any one part from each question (3 Marks x 4 Questions = 12 Marks)				
1.(a)	Derive the expression of torque developed in a DC motor. Discuss how we can reverse direction of rotation of a DC Motor	CO4	K2	1.2.1 1.4.1
1.(b)	Derive the expression of Generated EMF in DC Generators. How we can reverse the polarity of the brushes.	CO4	K2	1.2.1 1.4.1
2.(a)	A 4-pole DC generator with wave wound armature that has 51 slots, each slot having 24 conductors. The flux per pole is 0.01 Wb. At what speed the armature must be rotate to develop an induced emf of 220V.	CO4	K3	1.4.1 2.1.3
2.(b)	A 3-phase, 50 Hz, Induction motor has a full load speed of 1460 rpm. Calculate Slip, No. of poles, and frequency of induced rotor EMF under running condition	CO4	K3	1.3.1 2.1.3
3.(a)	Explain the term Smart grid and differentiate between a traditional grid and smart grid.	CO5	K2	3.1.4 6.2.1
3.(b)	What is earthing? Why we need earthing? Differentiate between earthing and grounding.	CO5	K2	3.1.4 3.1.5
4.(a)	With the help of examples, discuss the role and applications of Artificial Intelligence (AI) in modern power systems.	CO5	K2	5.1.1 5.1.2
4.(b)	What is a Battery Management System (BMS)? With the help of block diagram discuss working of a BMS.	CO5	K2	5.1.1 5.1.2

SECTION-B (Knowledge Level: K3)

Attempt any one part from each question (4 Marks x 2 Questions = 8 Marks)

5.(a)	Why 1- ϕ induction motor is not self-starting? Explain any one method of starting.	CO4	K2	1.2.1 1.4.1
5.(b)	Explain the working principle of a three-phase induction motor. Also define slip of an induction motor.	CO4	K2	2.1.3 2.1.4
6.(a)	A battery is discharged at a constant current of 1.2 A for 12 hours with average terminal voltage being 1.44 V. For charging, a steady current of 3.2 A for 5 hours is required with average terminal voltage being 1.66 V. Calculate the ampere-hour and watt-hour efficiency of the given battery.	CO5	K3	2.1.2 2.1.3
6.(b)	Calculate the Battery backup for a camping setup that uses a 24V, 100Ah lithium-ion battery which supplies a load consisting of a LED light of 20W, a small refrigerator of 40W and a laptop charger of 60W.	CO5	K3	1.3.1 2.1.2 2.1.3

CO Course Outcomes mapped with respective question KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create.

PI
arks)
6.1.1
7.1.1
6.1.1
2.2.3
12.2.1
8.2.2
11.3.2
6.2.1
arks)
11.1.1
2.2.3
6.2.1
11.3.2